

LOT: 一个面向工业的本体工程框架 LOT: An Industrial Oriented Ontology Engineering Framework

核心内容

Q: LOT 想解决什么问题? A: 本体工程领域这几十年出了一大堆方法论、指南、工具, 但真到开一个新项目时, 还是有很多没人回答的问题: 整个项目怎么管? 一个活动做完怎么接下一个? 每一步该用什么工具? LOT 就是把这些串成一条能照着走的路。它的定位是「轻量」——不是又造一套理论, 是把已有方法论里能落地的部分挑出来, 按软件行业的做法重新排一遍。

Q: 它的主流程是什么? A: 四步, 反复迭代: 需求规格 → 实现 → 发布 → 维护。前三步的骨架来自已有方法论里那条常见的「需求-实现-评估」链, 第四步「发布」是它加的, 此前的方法论都没把发布当成一项正式活动。

Q: 它和 METHONTOLOGY、NeOn 那些老方法论的关键区别在哪? A: 三处。一、需求不强制写成能力问题, 可以写成自然语言陈述, 也可以写成表格, 三种混着用; 二、和软件开发对齐——按冲刺迭代、引入持续集成的概念、版本约定照抄软件那一套; 三、发布是一等活动, 要有人读的文档、有机器读的文件、有能解析的地址。

Q: 为什么不强制用能力问题? 这是全文最实用的一条判断。 A: 作者从二十多年经验里得出的观察是: 能力问题这个技法在本体工程师、尤其是熟悉查询语言的人当中更常用, 但领域专家常常不熟悉那一套, 于是他们提的问题会偏泛、和数据实际怎么被查关系不大。更要命的是有些知识天生就是陈述句, 硬掰成问句很别扭——作者举的例子是「一个传感器观测某个属性」被迫写成「什么被传感器观测?」。这时候自然语言陈述更顺。而如果这个领域已经有结构化资料, 表格是最实用的, 因为它是领域和模型之间的中间形态。

Q: 复用分成硬的和软的, 差别是什么? A: 硬复用是把别人的本体整个导入进来, 用 OWL 的 imports 语句, 而且是传递的——被导入那份自己导入的东西也会一起进来。软复用只是引用对方那些元素的标识符, 有时会顺带把对方的一部分抄进自己这边。作者引的实证研究说: 实践中大多数所谓复用其实是软复用, 不是导入。

Q: 它对「什么时候去找可复用的本体」给了一条明确的劝阻, 是什么? A: 不要从需求出发去穷举地找现成本体。原话是那样做「非常费时」——从需求里的词去搜, 会搜出一大堆模型要研究、理解、比较。正确顺序是先做出一个初步的概念化, 等它稳定了、但还留有修改余地时, 再去找可以复用的。

Q: 评估分成两件事, 哪两件? A: 验证 (validation) 是拿本体的定义去对照你想描述的那个真实世界, 看意思对不对; 核验 (verification) 是拿本体去对照需求规格文档, 看是不是按规矩建的。前者问「建对了吗」, 后者问「按说好的建了吗」。作者还给了一条把需求变成测试用例的做法: 每个测试带上关联需求的编号、测试说明 (含指回需求文档的链接)、以及从能力问题抽出来的查询语句加预期结果。这样需求在整个开发过程里是可追溯的。

背景补充 / 点评

先说这份东西的分量。三位主要作者来自马德里理工大学本体工程组，那是欧洲本体工程的老牌团队之一。这篇不是提出一个新想法，是把二十多年、十八个真实项目的经验固化成流程——文中反复出现的「Tip」小节，写的就是「我们实际做下来发现」。发表在《人工智能的工程应用》，被引 169 次。

它的自我定位很清楚：把本体开发变成软件行业的一部分。这句话是摘要里的原话。所以它才会去谈冲刺、版本号约定、持续集成、议题追踪系统——这些在传统本体方法论里几乎不提。

我认为它最有用的不是那四步流程，是那些「Tip」。流程图谁都会画，难的是「什么时候该做什么」这类只有踩过坑才知道的判断。比如：概念化可以先在纸上或白板上画、拍照存档；用支持多人同时编辑的画图工具，把演进中的图当作决策记录；文档里的「使用示例」应该等评估完再写，因为评估之后模型常常还要改。

留白最明显的一处是评估。它把评估拆成验证和核验，也给了测试用例的形态，但对「怎么判断领域覆盖够不够」几乎没有可操作的办法，只是指向了另外两篇综述。这一处是全篇最虚的。

对我本人的启示，四条。

第一条要先纠正我自己。我们现在用的是 SAMOD 的方法论加 METHONTOLOGY 的需求规格模板拼起来的，而这两个都不是为工业项目写的。LOT 恰恰是。但我不认为要换掉现在这套——LOT 的四步里，「发布」那一步对我们不成立（我们的本体不对外发布），而它的需求规格模板本身也是从 NeOn 改的、和我们抄的那个同源。真正该抄的是它的 Tip 和它的复用分类，不是它的流程。

第二条最直接：需求不必全写成能力问题。我们的 CQ 注册表现在是唯一形态，所有需求都被掰成问句。LOT 说的那个「硬掰很别扭」的情况，我们正在发生——比如「一个考试属于一个小节」这种结构性知识，写成能力问题就是「一个考试属于哪个小节？」，这个问句没人会真的问，它只是为了塞进 CQ 格式而存在。LOT 给的三选一（能力问题、自然语言陈述、表格）值得考虑引入，尤其是表格那一种，因为我们本来就有结构化的物理表清单。

第三条是硬复用与软复用这个区分，它正好命中我们的一处设计。我们在学员画像那份设计里定了一条「临时正本加标识符对齐」的降级路径——课程侧的能力体系不存在时，先自己登一份带稳定标识符的，等对方就绪后做标识符映射。这在 LOT 的话里就是软复用，而且作者引的实证说这才是实践中的常态。我们当时是凭工程直觉定的，现在有了名字和出处。

第四条是一个警告。LOT 明确劝阻「从需求出发穷举地找现成本体」，理由是费时。我们这几周做的正是这件事——先把 GUMO、SUMO、SPOnto、OntoKG、UCPO 全读完，再回头设计。但我判断我们没错，因为 LOT 那条劝阻的前提是「你已经知道自己要建什么」，而我们当时连画像该有哪些字段都没定，读文献本身就是在做概念化。这一条记下来，下一个本体项目开工时用得上：知道要建什么了，就别再遍历文献。

文中金句

"This methodology aims to be compatible with software development techniques in which sprints and iterations represent the main workflow organization in order to align the ontology development with software development agile practices." 这套方法论力求与软件开发技术兼容——在那里，冲刺和迭代是主要的工作流组织方式——从而让本体开发与软件开发的敏捷实践对齐。

"In other cases, it might be unnatural or forced to reformulate as a question knowledge descriptions that are usually expressed as statements, for example transforming 'A sensor observes a property' to a question like 'What is observed by a sensor?'" 另一些情况下，把那些通常以陈述句表达的知识描述硬改写成问句，会显得不自然、很勉强，比如把「一个传感器观测某个属性」改成「什么被传感器观测？」这样的问题。

"ontology reuse is often not performed by means of importing the reused ontologies (hard reuse), but just by referencing the URIs of the reused ontology elements (soft reuse)." 本体复用往往不是通过导入被复用的本体来实现的（硬复用），而只是引用被复用本体中那些元素的标识符（软复用）。

"we discourage looking for ontologies exhaustively from the ontology requirements without having a more clear idea of the conceptualization, as it can be very time consuming to study, understand and compare the many models that can arise from searching from the terms in the requirements." 我们不建议在还没有一个较清晰的概念化想法时，就从本体需求出发去穷举地寻找现成本体——因为从需求里的词去搜会冒出很多模型，研究、理解、比较它们会非常费时。

全文翻译

摘要

Ontology Engineering has captured much attention during the last decades leading to the proliferation of numerous works regarding methodologies, guidelines, tools, resources, etc. including topics which are still being investigated. Even though, there are still many open questions when addressing a new ontology development project, regarding how to manage the overall project and articulate transitions between activities or which tasks and tools are recommended for each step.

本体工程在过去几十年里受到大量关注，催生了众多关于方法论、指南、工具、资源等等的工作，其中一些主题至今仍在研究之中。尽管如此，当着手一个新的本体开发项目时，仍有许多悬而未决的问题：整个项目该怎

么管理、各项活动之间的过渡该怎么衔接、每一步推荐用哪些任务和工具。

In this work we propose the Linked Open Terms (LOT) methodology, an overall and lightweight methodology for building ontologies based on existing methodologies and oriented to semantic web developments and technologies. The LOT methodology focuses on the alignment with industrial development, in addition to academic and research projects, and software development, that is making ontology development part of the software industry. This methodology includes lessons learnt from more than 20 years in ontological engineering and its application on 18 projects is reported.

本工作提出「链接开放术语」方法论（LOT），这是一套建立在既有方法论之上、面向语义网开发与技术的、整体而轻量的本体构建方法论。[LOT 方法论着重于与工业开发的对齐](#)（除了学术与研究项目之外），以及与软件开发的对齐——**也就是让本体开发成为软件产业的一部分**。这套方法论包含了本体工程二十多年的经验教训，并报告了它在 18 个项目中的应用。

一、引言：它想回答的八个问题

Q1: Should you always write ontology functional requirements in the form of Competency Questions (CQs) when you elaborate the requirements specification for an ontology? Q2: Which is the best way to communicate requirements to software engineers? Q3: And to domain experts? Q4: Which are the main problems that you will find when you reuse an ontology? Q5: Which is the sequence of activities that has been shown to be successful in practice in ontology development? Q6: What tools can you use for each activity in ontology development? Q7: Are there some tools that aid you to identify typical mistakes in modeling? Q8: What do you have to do to transform your Web Ontology Language (OWL) ontology into an HTML document?

问题一：在编写本体的需求规格时，本体的功能需求是否总该写成能力问题的形式？**问题二：**把需求传达给软件工程师，最好的方式是什么？**问题三：**传达给领域专家呢？**问题四：**复用一个本体时，你会遇到的主要问题有哪些？**问题五：**在本体开发实践中，被证明行之有效的活动顺序是什么？**问题六：**本体开发的每项活动可以用什么工具？**问题七：**有没有工具能帮你发现建模中的典型错误？**问题八：**要把你的 OWL 本体转成一份 HTML 文档，你得做什么？

二、方法论总览

The Linked Open Terms (LOT) methodology is a lightweight methodology for developing ontologies and vocabularies focusing on industry projects. This methodology aims to be compatible with software development techniques in which sprints and iterations represent the main workflow organization in order to align the ontology development with software development agile practices.

LOT 方法论是一套用于开发本体与词表的轻量方法论，聚焦于工业项目。[这套方法论力求与软件开发技术兼容](#)——在那里，冲刺和迭代是主要的工作流组织方式——从而让本体开发与软件开发的敏捷实践对齐。

In addition, the methodology focuses on: (a) the reuse on terms (ontology classes, properties and attributes) existing in already published vocabularies or ontologies and (b) on the publication of the built ontology according to Linked Data principles.

此外，这套方法论着重于两件事：（甲）**复用已发布词表或本体中已有的术语**（本体的类、属性与特征）；（乙）**按链接数据的原则发布所建成的本体**。

The LOT methodology defines iterations over a basic workflow composed of the following activities: (1) Ontological requirements specification; (2) Ontology implementation; (3) Ontology publication; and (4) Ontology maintenance. This main workflow has been enriched with Semantic Web oriented best practices and goals as the ontology publication, activity not taken into account in previous methodologies.

LOT 方法论在一条基本工作流上定义迭代，这条工作流由以下活动构成：**一、本体需求规格；二、本体实现；三、本体发布；四、本体维护**。这条主工作流在既有方法论的基础上，加入了面向语义网的最佳实践与目标，**比如本体发布——这项活动是此前的方法论都没有考虑过的**。

The LOT methodology also combines and offers different ways of describing ontology requirements (Competency Questions, Natural language statements and tabular information based on METHONTOLOGY) instead of a fix or unique technique. It also provides a template for ontology functional requirements description including information to ease traceability.

LOT 方法论还组合并提供了描述本体需求的**多种不同方式**（能力问题、自然语言陈述、基于 METHONTOLOGY 的表格信息），**而不是固定或唯一的一种技法**。它同时提供了一份本体功能需求描述的模板，其中包含便于追溯的信息。

Another novelty of the LOT methodology is the alignment with software development practices and tools as well as the inclusion of the continuous integration notion in the ontology developments. Finally, the LOT methodology include tips and recommendations based on more than 20 years of experience for each activity.

LOT 方法论的另一个新意，在于与软件开发实践和工具的对齐，以及**把持续集成的概念引入本体开发**。最后，LOT 方法论为每一项活动都给出了基于二十多年经验的提示与建议。

三、第一步：本体需求规格

The ontology requirements specification activity refers to the activity of collecting the requirements that the ontology should fulfill. These requirements are usually related to the goal of the ontology, to the domain that the ontology should model, or to technical details of the ontology like the implementation language, among others. In this step, involvement and commitment by experts in the specific domain at hand is required to generate the appropriate industry perspective and knowledge.

本体需求规格这项活动，指的是收集这个本体应当满足的需求。这些需求通常与本体的目标、本体要建模的领域、或本体的技术细节（比如实现语言）等等相关。**在这一步，需要相关领域专家的参与和投入**，才能产生恰当的行业视角与知识。

3.1 用例规格

The goal of the use case specification activity is to provide a vision on the potential use that the ontology will have. This activity involves domain experts, users and ontology engineers. The specification of a use case consists of the narration of a scenario in which one or several actors intervene and the necessary interactions to obtain an expected result or benefit.

用例规格这项活动的目标，是勾勒出这个本体潜在的用途图景。这项活动涉及领域专家、用户和本体工程师。**一个用例的规格由一段场景叙述构成**，其中有一个或多个角色参与，以及为获得预期结果或收益所需的那些交互。

3.2 资料交接

The goal of the data exchange identification activity is to provide the ontology development team with the necessary documentation about the domain to be modeled. In this case, the documentation to be shared might correspond to: datasets, regulations, standards, data formats, APIs specifications, database schemas, etc. Typically in this activity, ontology users such as software developers are responsible for providing this documentation to ontology engineers.

资料交接这项活动的目标，是把待建模领域的必要文档交到本体开发团队手里。这里要共享的文档可能包括：**数据集、法规、标准、数据格式、接口规格、数据库表结构**等等。这项活动中，通常由本体的使用者（比如软件开发者）负责把这些文档提供给本体工程师。

3.3 功能需求：三种写法，不强制用能力问题

While traditionally competency questions have attracted attention from many methodologies, LOT proposes to use also other techniques depending on the case at hand or to combine them. In general, the competency questions and natural language statements are commonly used in combination.

尽管传统上能力问题一直是众多方法论关注的重点，**LOT 主张也可以根据具体情况使用其他技法，或者把它们组合起来**。一般而言，能力问题与自然语言陈述常常是配合使用的。

From our experience, it has been observed that using the competency questions technique is more common among ontology engineers or teams with experience in query languages, mostly SPARQL. However, since domain experts are sometimes not familiar with the OWL and SPARQL capabilities, they could tend to generate more general questions not so much related on how the data is going to be queried.

根据我们的经验，我们观察到：能力问题这个技法在**本体工程师、或有查询语言（多半是 SPARQL）经验的团队当中更常用**。然而，由于领域专家有时并不熟悉 OWL 与 SPARQL 的能力，他们往往会提出更笼统的问题，与数据实际将被如何查询关系不大。

In other cases, it might be unnatural or forced to reformulate as a question knowledge descriptions that are usually expressed as statements, for example transforming "A sensor observes a property" to a question like "What is observed by a sensor?" In these cases the natural language statements help. Also, if there is already structured information about the domain, tables are also useful and more practical for generating a first conceptualization, because they serve as an intermediate representation between the domain and the model.

另一些情况下，把那些通常以陈述句表达的知识描述硬改写成问句，会显得不自然、很勉强，比如把「一个传感器观测某个属性」改成「什么被传感器观测？」这样的问题。这种时候自然语言陈述就派上用场了。此外，如果这个领域已经有结构化的信息，表格也很有用、而且对生成第一版概念化更实际，因为它们充当了领域与模型之间的中间表示。

3.4 需求补全与优先级

Domain experts and users in collaboration with the ontology development team validate whether the ontology requirements defined in the previous step are valid and complete. If needed, functional requirements are assigned priorities. The main implication of this prioritization is the possibility of planning and scheduling the development of the ontology in sprints.

领域专家和用户与本体开发团队协作，验证上一步定义的本体需求是否有效、是否完整。**如有需要，功能需求会被赋予优先级**。这种优先级排序的主要意义在于：**使得按冲刺来规划和排期本体开发成为可能**。

3.5 需求形式化成测试用例

In this (optional) task, the functional requirements are formalized into test cases. These tests cases should include: the identifier of the requirement associated, the description of the test case (which includes a link to the ORSD), and the SPARQL queries extracted from the CQ together with the expected result of the query. The definition and execution of these tests allow the requirements traceability during the ontology development process.

在这项（可选的）任务中，功能需求被形式化成测试用例。**这些测试用例应当包含：所关联需求的标识符、测试用例的说明（其中含有指向需求规格文档的链接）、以及从能力问题中抽出的 SPARQL 查询连同该查询的预期结果**。定义并执行这些测试，使得需求在整个本体开发过程中是可追溯的。

四、第二步：本体实现

The ontology development team builds the ontology iteratively, implementing only a certain number of requirements in each iteration. The output of each iteration is a new version of the ontology.

本体开发团队**迭代式地**构建本体，每次迭代只实现一定数量的需求。每次迭代的产出是本体的一个新版本。

4.1 概念化

The purpose of the conceptualization sub-activity is to create a model that represents the domain of the ontology. Such conceptualization usually does not include all the constraints that a formal language imposes; instead the level of detail is decided by the development team.

概念化这项子活动的目的，是创建一个能代表本体所涉领域的模型。**这样的概念化通常不包含形式语言强加的全部约束**，详细程度由开发团队自己决定。

When using diagrams, either on paper or using a diagramming tool, it is important to decide the notation to be used as it will be part of the common understanding between domain experts and ontology developers and it will also drive the future ontology encoding.

使用图示时——无论是画在纸上还是用绘图工具——**决定采用哪种画法很重要**，因为它会成为领域专家与本体开发者之间共同理解的一部分，也会驱动后续的本体编码。

It is important to take into account that the conceptualization activity could be carried out on paper or on blackboard from where photographs of the conceptualization diagrams are taken to generate digital drawings and keep track of the conceptualization results. Another option is to use digital drawing tools, ideally with synchronous collaborative edition capabilities, and to use the evolving diagrams to keep history of the model and track decisions taken.

（提示）值得注意的是，**概念化活动完全可以在纸上或黑板上进行，然后把概念图拍下来**，据此生成电子图并留存概念化的结果。另一个选项是使用电子绘图工具，**最好是支持多人同步协作编辑的**，并用不断演进的图来保存模型的历史、追踪做过的决策。

4.2 编码与版本

Even though versioning conventions should be established for all artefacts being produced during the project, this practice has not yet been broadly adopted in ontology development projects. Therefore, it is advisable to establish a versioning convention at least to identify the different ontology versions produced along the project. For doing so, the ontology development team should define a versioning convention taking as inspiration the conventions used in software development.

尽管项目过程中产出的所有产物都应当建立版本约定，**这个做法在本体开发项目里还没有被广泛采纳**。因此，建议至少建立一套版本约定来标识项目过程中产出的不同本体版本。为此，**本体开发团队应当参照软件开发中使用的约定来定义自己的版本约定**。

4.3 复用：硬的与软的

As presented in Poveda-Villalón et al. (2012), ontology reuse is often not performed by means of importing the reused ontologies (hard reuse), but just by referencing the URIs of the reused ontology elements (soft reuse).

如 Poveda-Villalón 等人 2012 年的工作所指出的，**本体复用往往不是通过导入被复用的本体来实现的（硬复用），而只是引用被复用本体中那些元素的标识符（软复用）。**

Hard reuse: This type of reuse is implemented by the OWL construct owl:imports. In this case the reused ontology is reused as it is, and as a whole. As the owl:imports construct is transitive, the imported ontologies by the one being reused will be also part of the ontology being built.

硬复用：这种复用通过 OWL 的 imports 构造来实现。这种情况下，被复用的本体是**原样地、整体地**被复用。由于 imports 构造是传递的，**被复用那份自己所导入的本体，也会成为正在构建的这个本体的一部分。**

Soft reuse: This type of reuse is implemented by referring to other ontology elements URIs. It could also include the reproduction of some parts of the reused ontology in the reusing one.

软复用：这种复用通过引用其他本体元素的标识符来实现。它也可以包括把被复用本体的某些部分复制到复用方这一侧。

It should be taken into account that even though there are many benefits claimed about ontology reuse, such an activity is not cost free and some common issues might arise: heterogeneity in the natural language used, heterogeneity between the concepts of the reusing and the reused ontologies, deficiencies in the documentation, loss of information due to an imported ontology that is not available, unavailable license, etc.

应当注意，尽管人们宣称本体复用有诸多好处，**这项活动并非没有成本**，一些常见问题可能会冒出来：**所用自然语言的异质性、复用方与被复用方概念之间的异质性、文档的不足、因被导入的本体不可访问而丢失信息、许可证不可用等等。**

The ontology reuse activity might be carried out in line with the ontology encoding one, that is, once the model is stable, look for ontologies that already represent such model. From our experience, we discourage looking for ontologies exhaustively from the ontology requirements without having a more clear idea of the conceptualization, as it can be very time consuming to study, understand and compare the many models that can arise from searching from the terms in the requirements.

（提示）本体复用活动可以与本体编码活动同步进行，也就是说，**一旦模型稳定下来，再去寻找已经表达了该模型的本体。**根据我们的经验，**我们不建议在还没有一个较清晰的概念化想法时，就从本体需求出发去穷举地寻找现成本体——因为从需求里的词去搜会冒出很多模型，研究、理解、比较它们会非常费时。**

The most convenient situation is starting the conceptualization without looking for many ontologies (but considering those that the development team is knowledgeable about) and

looking for ontologies once a first conceptualization is stable but open to modifications according to the ontologies to be reused.

最合适的做法是：先开始概念化，不去大量搜寻本体（但把开发团队本来就熟悉的那些考虑进来），**等第一版概念化稳定下来、但仍对修改保持开放时，再去寻找可复用的本体。**

4.4 评估：验证与核验是两件事

Evaluation can be divided into two categories: (1) validation, which compares the meaning of the ontology definitions against the intended model of the world aiming to conceptualize; and (2) verification, which compares the ontology against the ontology specification document (ontology requirements and competency questions), ensuring that the ontology is built correctly.

评估可以分成两类。**一、验证**：把本体定义的含义，与你想要概念化的那个世界的预期模型作比较。**二、核验**：把本体与本体规格文档（本体需求与能力问题）作比较，确保本体是被正确地构建的。

From our experience, it has been observed that when there are domain experts involved in the ontology development process and the ontology is not a part of a software system, the execution of SPARQL queries or test expressions in order to gather data based on ontology requirements helps domain experts to easily identify misunderstandings and lack of knowledge described in an ontology.

（提示）根据我们的经验，我们观察到：**当有领域专家参与本体开发过程、而这个本体又不是某个软件系统的一部分时，按本体需求执行 SPARQL 查询或测试表达式来取数，能帮助领域专家轻松地发现本体中描述的误解与知识缺失。**

五、第三步：发布——此前方法论都没有的一环

The aim of the ontology publication activity is to provide an online ontology accessible both as a human-readable documentation and a machine-readable file from its URI.

本体发布这项活动的目的，是提供一个在线的本体，**从它的标识地址既能取到人可读的文档，也能取到机器可读的文件。**

Once the ontology developers have implemented and evaluated the ontology, the next task is to decide whether the current version is going to be published on the Web or whether it is needed to document such versions of the ontology for any other reason. In this case, the version at hand becomes a release candidate.

一旦本体开发者实现并评估完本体，下一项任务就是决定：当前版本是否要发布到网上，或者是否出于其他原因需要为这些版本编写文档。**这种情况下，手上的这个版本就成为一个候选发布版。**

From our experience, we observed that it is very useful to include within the HTML documentation graphical representations of ontology instantiations and examples of use,

together with the ontology diagrams. Note that while one might be tempted to generate as much documentation as possible during the conceptualization and implementation, in some cases a number of modifications are needed after evaluating the ontology from a functional point of view. In this sense, it is recommended to accomplish some parts of the documentation (for example the examples generation) after the evaluation activity.

（提示）根据我们的经验，我们观察到：[在 HTML 文档里放上本体实例化的图形表示和使用示例、连同本体的图，非常有用](#)。需要注意的是，虽然人们会忍不住在概念化和实现阶段就尽可能多地生成文档，[但从功能角度评估完本体之后，往往还需要做不少修改](#)。因此，[建议文档的某些部分（比如示例的生成）放到评估活动之后再](#)做。

六、第四步：维护

The goal of this activity is to update the ontology during its life cycle. These situations are bug detection, which also include suggestions and improvements, and new requirements identification. In both cases, the involved actor (developer, user, etc.) detecting the bug or requirement should report it to the ontology development team.

这项活动的目标是在本体的生命周期里更新它。[触发它的情形有两种：发现缺陷](#)（也包括建议与改进），[以及识别出新需求](#)。两种情况下，发现缺陷或需求的那一方（开发者、用户等等）都应当把它报告给本体开发团队。

Monitoring and maintaining the traceability of the requirements, bugs and suggestions identified in this activity allows storing discussions and decisions taken that can be later reused. For this reason, it is advisable to use issue tracker systems that allow issue management and discussion, possibly in an open environment so that external feedback can be received.

（提示）[监控并维持这项活动中所识别的需求、缺陷与建议的可追溯性，能把讨论过程和做过的决策保存下来，供日后复用](#)。出于这个原因，建议使用支持议题管理与讨论的议题追踪系统，[最好是在开放环境中，以便能收到外部反馈](#)。

参考链接

- [LOT 方法论在线文档](#) — 方法论本身的官方站点，含各活动的详细说明
- [LOT 资源仓库](#) — 需求规格文档模板（docx 与 tex 两种格式），从 NeOn 方法论的模板改编而来
- [CORAL 语料库](#) — 从真实本体中抽取的 834 条需求，按词法句法模式标注；含一部模式词典，可用来识别需求的类型和常见歧义表达
- [Chowlk 画法](#) — 一套基于 UML 的本体图示画法，用于概念化阶段
- [本体设计模式门户](#) — 数十个领域的可复用建模模式目录
- [核验测试用例本体](#) — 把本体需求和测试用例存成 RDF 文件的词表

- [W3C 多元关系建模说明](#) — 关于怎么给多元关系建模的官方指引